

Hyun-Joon Yang

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EDUCATION

University of Southern California

B.S., Computational Neuroscience | Minor in Computer Science

August 2015 – May 2021

GPA: 3.784

RESEARCH EXPERIENCE

Precision Psychiatry Data Analyst – Clinical & Behavioral Data Lead

September 2025 – Present

PanLab @ Department of Psychiatry, Stanford School of Medicine

PI: Dr. Leanne Williams

- Promoted to lead the management, integration, analysis, and interpretation of clinical and behavioral data
- Developed a Python package for processing, compiling, and harmonizing psychometric datasets
- Created visualizations and conducted preliminary analyses to support ongoing research objectives
- Automated data processing workflows, enhancing efficiency in data handling and analysis

Precision Psychiatry and Neuroimaging Data Analyst

July 2024 – September 2025

PanLab @ Department of Psychiatry, Stanford School of Medicine

PI: Dr. Leanne Williams

- Led coordination with the DCC to process and submit data, in compliance with NIMH reporting requirements for U01-funded clinical trial
- Spearheaded collaboration between engineers and clinical research teams to develop, test, and launch a web-based cognitive battery, integrating it into the study
- Harmonized multiple archival datasets for machine learning analysis
- Validated procedures for scoring behavioral data based on normative samples, integral to the lab's research
- Presented a novel offline policy evaluation method to demonstrate that using brain signatures for treatment assignment results in higher remission rates compared to traditional methods

Research Data Analyst

June 2021 – July 2024

Department of Psychiatry, Stanford School of Medicine

PI: Dr. Jane Paik Kim

- Supported research activities in the Empirical Ethics Group, Adolescent Eating Disorder Program, and the Center for Interdisciplinary Brain Sciences Research
- Developed and deployed a survey administration web application using Python Flask and AWS (e.g., EC2, BeanStalk), implementing the Sequential Multiple Assignment Randomized Trials (SMART) methodology
- Applied advanced statistical algorithms (e.g., Graphical Lasso, Latent Dirichlet Allocation, Contextual Bandits) to analyze the ethical implications of ML/AI in medical decision-making and neuroscience research participation
- Designed a dashboard that integrates data from a mobile health app to deliver insights for clinicians on adolescent anorexia nervosa (AN)

Research Assistant

May 2019 – Dec 2020

Microstructural Imaging Group @ Laboratory of NeuroImaging (LONI)

PI: Dr. Farshid Sepehrband

- Created a Deep Neural Networks model on a GPU cluster that auto-segments White Matter Hyper-intensities in multi-modal MRI images of the brain with Dice coefficient of up to 0.88

PROJECTS

Brain Metastases Segmentation

UNet, PyTorch, SLURM

- Trained a UNet using PyTorch on an HPC cluster to segment metastases in multi-modal brain images

Housing Price Prediction

MLP, RF, XGBoost, AdaBoost, Ridge Regression

- Employed various ML algorithms to predict housing prices and achieve 0.116 RMSE

Transcription Factor Binding Specificity

R, Linear Regression

- Incorporated DNA shape information in regression models to evaluate prediction of TF binding specificity in R

Image Registration

Linear Algebra, Gradient Descent

- Demonstrated knowledge of math behind image registration as part of a take-home technical challenge

CollabDraw

WebSocket, Flask

- Developed a web application for real-time collaborative doodling

Rate My Dorm

PHP, Bootstrap, MySQL

- Built a full-stack web app for rating and reviewing USC dorms, integrating user authentication and dynamic content

COVID-19 Dashboard

Flask, MySQL, Linear Regression

- Created a web dashboard for trajectories of COVID-19 cases in the US in real-time based on a regression model

TECHNICAL SKILLS

Languages: C/C++, Python, MATLAB, R, Javascript, HTML, CSS, PHP, LaTeX

Libraries: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Flask, Dash

Tools: Git, Conda, UNIX/Linux, Jupyter Notebook, HPC/SLURM, Docker

Databases: MongoDB, MySQL, REDCap

CONFERENCE PAPERS

- [C1] GiSoo Kim, Jane P Kim, and **Hyun-Joon Yang**. “Robust Tests in Online Decision-Making”. In: *Proceedings of the 36th AAAI Conference on Artificial Intelligence*. 2022.
- [C2] Farshid Sepehrband, Giuseppe Barisano, **Hyun-Joon Yang**, Jeiran Choupan, and Arthur W Toga. “WMH and PVS mapping from clinical data using semi-supervised multi-modal convolutional neural network”. In: *Proceedings of the 2020 ISMRM & SMRT Virtual Conference & Exhibition*. 2020.

JOURNAL ARTICLES

- [A1] Christina Miranda, Brittany Matheson, Nandini Datta, Aileen Whyte, **Hyun-Joon Yang**, Paul Schmiedmayer, Vishnu Ravi, Oliver Aalami, and James Lock. “Enhancing Distress Tolerance Skills in Adolescents With Anorexia Nervosa Through the BALANCE Mobile App: Feasibility and Acceptability Study”. In: *JMIR Formative Research* (2025).
- [A2] Katie Ryan, **Hyun-Joon Yang**, Bohye Kim, and Jane Paik Kim. “Assessing the impact of AI on physician decision-making for mental health treatment in primary care”. In: *npj Mental Health Research* (2025).
- [A3] Jane Paik Kim and **Hyun-Joon Yang**. “A novel experimental vignette methodology: SMART vignettes”. In: *Methodological Innovations* (2024).
- [A4] Jane Paik Kim, **Hyun-Joon Yang**, Bohye Kim, Katie Ryan, and Laura Weiss Roberts. “Understanding Physician’s Perspectives on AI in Health Care: Protocol for a Sequential Multiple Assignment Randomized Vignette Study”. In: *JMIR Research Protocols* (2024).
- [A5] Brittany Matheson, Nandini Datta, Eliza Van Wye, **Hyun-Joon Yang**, and James Lock. “Pilot Case Series Studying a Psychoeducational and Motivational Treatment for Children With Low-Weight Avoidant Restrictive Food Intake Disorder”. In: *International Journal of Eating Disorders* (2024).
- [A6] Eliza Van Wye, Brittany Matheson, Kyra Citron, **Hyun-Joon Yang**, Nandini Datta, Cara Bohon, and James D. Lock. “Protocol for a randomized clinical trial for Avoidant Restrictive Food Intake Disorder (ARFID) in low-weight youth”. In: *Contemporary Clinical Trials* (2023).
- [A7] Kyra Citron, Madelyn Johnson, Brittany E. Matheson, Z. Ayotola Onipede, **Hyun-Joon Yang**, Cara Bohon, Daniel Le Grange, and James Lock. “Study protocol for training providers in private practice in family-based treatment for adolescents with anorexia nervosa: A randomized controlled feasibility trial”. In: *Contemporary Clinical Trials* (2022).